

DD_PLM-CFG-07 - Voice Tariff Catalog - Developer Implementation Guide

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1. How To Read This Guide

This guide follows the flow a developer should implement.

The catalog does not rate CDRs. It only stores and exposes tariff data. RAT-01 calls the lookup API and performs the charge calculation.

2. Step 1 - Create Destination Zones

Start by creating the zones that rating will use.

Example zones for Kenya launch:

```
insert into voice_destination_zone (
  zone_id,
  operator_code,
  zone_code,
  zone_name,
  zone_type,
  default_charge_policy,
  status,
  created_at,
  updated_at
) values
('vdz-wik-onnet', 'WIK', 'WIK_ON_NET', 'Wananchi On-net', 'ON_NET', 'CHARGEABLE', 'ACTIVE', now(), now()),
('vdz-ke-mobile', 'WIK', 'KE_MOBILE', 'Kenya Mobile', 'NATIONAL', 'CHARGEABLE', 'ACTIVE', now(), now()),
('vdz-ke-fixed', 'WIK', 'KE_FIXED', 'Kenya Fixed', 'NATIONAL', 'CHARGEABLE', 'ACTIVE', now(), now()),
('vdz-emergency', 'WIK', 'EMERGENCY', 'Emergency Numbers', 'EMERGENCY', 'ZERO_RATED', 'ACTIVE', now(), now()),
('vdz-tollfree', 'WIK', 'TOLL_FREE', 'Toll Free', 'TOLL_FREE', 'ZERO_RATED', 'ACTIVE', now(), now()),
('vdz-international', 'WIK', 'INTERNATIONAL_ROW', 'International Rest of World', 'INTERNATIONAL', 'CHARGEABLE', 'ACTIVE', now(), now())
```

Meaning:

- zone_code is what RAT-01 stores on rated usage.
- zone_type helps reporting and validation.
- default_charge_policy protects special categories such as emergency calls.

3. Step 2 - Load Prefixes

Create prefixes that map normalized called numbers to zones.

```
insert into voice_destination_prefix (
  prefix_id,
  operator_code,
  prefix,
  zone_id,
  match_priority,
  status,
  effective_from,
  notes
) values
('vdp-ke-mobile-2547', 'WIK', '+2547', 'vdz-ke-mobile', 100, 'ACTIVE', '2026-06-01T00:00:00Z', 'Kenya mobile'),
('vdp-ke-fixed-25420', 'WIK', '+25420', 'vdz-ke-fixed', 100, 'ACTIVE', '2026-06-01T00:00:00Z', 'Nairobi fixed'),
('vdp-emergency-999', 'WIK', '999', 'vdz-emergency', 100, 'ACTIVE', '2026-06-01T00:00:00Z', 'Emergency'),
('vdp-emergency-112', 'WIK', '112', 'vdz-emergency', 100, 'ACTIVE', '2026-06-01T00:00:00Z', 'Emergency'),
('vdp-tollfree-0800', 'WIK', '0800', 'vdz-tollfree', 100, 'ACTIVE', '2026-06-01T00:00:00Z', 'Toll free');
```

3.1 Prefix Match Logic

Use longest-prefix match.

Pseudo-code:

```
public function resolveZone(string $operatorCode, string $calledNumber, Carbon $callStartedAt): ?DestinationZone
{
    return VoiceDestinationPrefix::query()
        ->where('operator_code', $operatorCode)
        ->where('status', 'ACTIVE')
        ->where('effective_from', '<=', $callStartedAt)
        ->where(function ($q) use ($callStartedAt) {
            $q->whereNull('effective_to')->orWhere('effective_to', '>', $callStartedAt);
        })
        ->whereRaw('? like prefix || \'%\''', [$calledNumber])
        ->orderByRaw('length(prefix) desc')
        ->orderByDesc('match_priority')
        ->first()?->zone;
}
```

If no prefix is found, return no zone. RAT -01 must quarantine the CDR instead of rating it.

4. Step 3 - Create Time Band

Start with ANYTIME if product does not need peak/off-peak.

```
insert into voice_time_band (
    time_band_id,
    operator_code,
    time_band_code,
    display_name,
    days_of_week,
    start_time_local,
    end_time_local,
    timezone,
    status
) values (
    'vtb-wik-anytime',
    'WIK',
    'ANYTIME',
    'Any time',
    'MON,TUE,WED,THU,FRI,SAT,SUN',
    '00:00:00',
    '23:59:59',
    'Africa/Nairobi',
    'ACTIVE'
);
```

Time band lookup must use the operator timezone, not server timezone.

5. Step 4 - Create Tariff Plan

Create the plan in DRAFT first.

```
insert into voice_tariff_plan (
    tariff_plan_id,
    operator_code,
    tariff_plan_code,
    display_name,
    billing_mode,
    currency_code,
    status,
    effective_from,
    created_at,
    updated_at
) values (
    'vtp-wik-res-voice-2026',
    'WIK',
    'WIK_RES_VOICE_STD_2026',
    'Residential Voice Standard 2026',
    'BOTH',
    'KES',
    'DRAFT',
    '2026-06-01T00:00:00Z',
    now(),
    now()
);
```

Keep plans in DRAFT until zones, prefixes, rates, bindings, and validation checks are complete.

6. Step 5 - Add Rates

Add rates per zone.

```
insert into voice_tariff_rate (
    rate_id,
    operator_code,
    tariff_plan_id,
    zone_id,
    time_band_id,
    call_direction,
    unit_type,
    unit_price,
    setup_fee_amount,
    initial_increment_seconds,
    subsequent_increment_seconds,
    minimum_charge_amount,
    charge_policy,
    taxable_kind,
    taxable_ref,
    effective_from,
    status
) values
('vtr-ke-mobile', 'WIK', 'vtp-wik-res-voice-2026', 'vdz-ke-mobile', 'vtb-wik-anytime', 'OUTBOUND',
'MINUTE', 3.000000, 0, 60, 60, 0, 'CHARGEABLE', 'SERVICE', 'svc_VOIP_STD', '2026-06-01T00:00:00Z', 'DRAFT'),
('vtr-ke-fixed', 'WIK', 'vtp-wik-res-voice-2026', 'vdz-ke-fixed', 'vtb-wik-anytime', 'OUTBOUND',
'MINUTE', 2.000000, 0, 60, 60, 0, 'CHARGEABLE', 'SERVICE', 'svc_VOIP_STD', '2026-06-01T00:00:00Z', 'DRAFT'),
('vtr-emergency', 'WIK', 'vtp-wik-res-voice-2026', 'vdz-emergency', 'vtb-wik-anytime', 'OUTBOUND',
'MINUTE', 0.000000, 0, 1, 1, 0, 'ZERO_RATED', 'SERVICE', 'svc_VOIP_STD', '2026-06-01T00:00:00Z', 'DRAFT'),
('vtr-tollfree', 'WIK', 'vtp-wik-res-voice-2026', 'vdz-tollfree', 'vtb-wik-anytime', 'OUTBOUND',
'MINUTE', 0.000000, 0, 1, 1, 0, 'ZERO_RATED', 'SERVICE', 'svc_VOIP_STD', '2026-06-01T00:00:00Z', 'DRAFT');
```

6.1 Validate Rate Overlap

Before activation, reject two active or draft rates that overlap for:

```
operator_code
tariff_plan_id
zone_id
time_band_id
call_direction
effective date range
```

Pseudo-code:

```
public function assertNoRateOverlap(VoiceTariffRate $candidate): void
{
    $overlap = VoiceTariffRate::query()
        ->where('operator_code', $candidate->operator_code)
        ->where('tariff_plan_id', $candidate->tariff_plan_id)
        ->where('zone_id', $candidate->zone_id)
        ->where('time_band_id', $candidate->time_band_id)
        ->where('call_direction', $candidate->call_direction)
        ->whereIn('status', ['DRAFT', 'ACTIVE'])
        ->where('rate_id', '<>', $candidate->rate_id)
        ->where('effective_from', '<', $candidate->effective_to ?? '9999-12-31')
        ->whereRaw('coalesce(effective_to, timestamp \'9999-12-31\') > ?', [$candidate->effective_from])
        ->exists();

    if ($overlap) {
        throw new DomainException('VOICE_RATE_OVERLAP');
    }
}
```

7. Step 6 - Add Allowance

If the plan has no included minutes, skip this step.

Example: 100 on-net minutes per bill cycle.

```
insert into voice_tariff_allowance (
    allowance_id,
    operator_code,
    tariff_plan_id,
    allowance_code,
    allowance_name,
```

```

        included_seconds,
        eligible_zone_codes_json,
        cycle_policy,
        carry_over_policy,
        priority,
        status,
        effective_from
    ) values (
        'vta-wik-onnet-100',
        'WIK',
        'vtp-wik-res-voice-2026',
        'ONNET_100_MIN',
        '100 on-net minutes',
        6000,
        ['WIK_ON_NET'],
        'BILL_CYCLE',
        'NONE',
        10,
        'ACTIVE',
        '2026-06-01T00:00:00Z'
    );

```

RAT-01 consumes the allowance. PLM-CFG-07 only defines it.

8. Step 7 - Bind Tariff Plan

Bind the tariff plan to a package or service.

Package-level binding:

```

insert into voice_tariff_binding (
    binding_id,
    operator_code,
    binding_scope,
    binding_ref,
    tariff_plan_id,
    priority,
    status,
    effective_from
) values (
    'vtbnd-pkg-voice-std',
    'WIK',
    'PACKAGE',
    'pkg_INET_VVOICE_STD',
    'vtp-wik-res-voice-2026',
    100,
    'ACTIVE',
    '2026-06-01T00:00:00Z'
);

```

Binding priority:

Binding scope	Priority rule
SUBSCRIPTION_OVERRIDE	Highest priority. Used for explicit customer exception.
PACKAGE	Normal product-level binding.
SERVICE	Fallback if package has no binding.

9. Step 8 - Activate Plan

Validation before activation:

1. Plan has at least one active/future rate.
2. Every chargeable active zone used by prefixes has at least one rate.
3. Emergency zone has zero-rated rate.
4. No rate overlaps.
5. Currency is valid for operator.
6. Binding references valid package/service/subscription.

Activation update:

```

update voice_tariff_plan
set status = 'ACTIVE',
    updated_at = now()
where tariff_plan_id = 'vtp-wik-res-voice-2026'
    and status = 'DRAFT';

update voice_tariff_rate
set status = 'ACTIVE'
where tariff_plan_id = 'vtp-wik-res-voice-2026'
    and status = 'DRAFT';

```

Emit:

```

{
  "eventType": "VoiceTariffPlanActivated",
  "operatorCode": "WIK",
  "tariffPlanCode": "WIK_RES_VOICE_STD_2026",
  "effectiveFrom": "2026-06-01T00:00:00Z"
}

```

RAT-01 consumes this event and refreshes cache.

10. Step 9 - RAT-01 Performs Lookup

When RAT-01 receives a mediated CDR, it calls:

```

POST /api/plm/voice-rating/lookup
Content-Type: application/json

{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "packageRef": "pkg_INET_VOICE_STD",
  "serviceRef": "svc_VOIP_STD",
  "calledNumberNormalized": "+254711222333",
  "callDirection": "OUTBOUND",
  "callStartedAt": "2026-06-02T10:15:00+03:00"
}

```

Lookup logic:

1. Resolve tariff plan by binding priority.
2. Resolve destination zone by longest-prefix match.
3. Resolve time band using operator timezone.
4. Resolve active rate by plan, zone, time band, direction, and call start time.
5. Return rate payload.

Response:

```

{
  "tariffPlanCode": "WIK_RES_VOICE_STD_2026",
  "zoneCode": "KE_MOBILE",
  "timeBandCode": "ANYTIME",
  "chargePolicy": "CHARGEABLE",
  "unitType": "MINUTE",
  "unitPrice": 3.0,
  "currency": "KES",
  "initialIncrementSeconds": 60,
  "subsequentIncrementSeconds": 60,
  "setupFeeAmount": 0.0,
  "minimumChargeAmount": 0.0,
  "taxableKind": "SERVICE",
  "taxableRef": "svc_VOIP_STD"
}

```

If no prefix or rate is found:

```

{
  "errorCode": "VOICE_TARIFF_NOT_FOUND",
  "reason": "No active destination prefix for +998..."
}

```

RAT-01 must quarantine the CDR.

11. Step 10 - Rating Calculation Example

PLM-CFG-07 response:

```
{
  "unitType": "MINUTE",
  "unitPrice": 3.0,
  "initialIncrementSeconds": 60,
  "subsequentIncrementSeconds": 60,
  "setupFeeAmount": 0.0,
  "minimumChargeAmount": 0.0
}
```

Mediated CDR:

```
{
  "durationSeconds": 75
}
```

RAT-01 calculates:

```
initial increment covers first 60 seconds
remaining actual seconds = 15
subsequent increment rounds remaining 15 to 60
billable seconds = 120
billable units = 120 / 60 = 2 minutes
charge = 2 * 3.00 + 0 setup fee = 6.00 KES
```

PLM-CFG-07 does not persist this calculation. RAT-01 persists rated usage.

12. Step 11 - Bulk Import Prefixes Or Rates

Backoffice uploads CSV to:

```
POST /api/plm/voice-destination-prefixes/bulk-import
```

Example CSV:

```
prefix,zone_code,effective_from,notes
+2547,KE_MOBILE,2026-06-01T00:00:00Z,Kenya mobile
+25420,KE_FIXED,2026-06-01T00:00:00Z,Nairobi fixed
999,EMERGENCY,2026-06-01T00:00:00Z,Emergency
```

Import rules:

1. Validate required columns.
2. Validate zone exists.
3. Validate no duplicate active prefix for same operator/effective date.
4. Insert in one batch transaction if file is small.
5. For large files, create import job and process asynchronously.
6. Emit prefix change event after success.

13. Error Codes

Error code	Meaning
VOICE_TARIFF_PLAN_NOT_FOUND	No applicable tariff plan binding.
VOICE_DESTINATION_ZONE_NOT_FOUND	Called number did not match a prefix.
VOICE_RATE_NOT_FOUND	Zone resolved but no active rate exists.
VOICE_RATE_OVERLAP	Rate effective periods overlap.
VOICE_PREFIX_DUPLICATE	Duplicate active prefix.
VOICE_BINDING_REF_INVALID	Package/service/subscription binding reference is invalid.
VOICE_TARIFF_BACKDATE_NOT_ALLOWED	User attempted backdated tariff change without permission.
VOICE_PREMIUM_RATE_REQUIRED	Premium zone exists without explicit active rate.
VOICE_EMERGENCY_ZERO_RATE_REQUIRED	Emergency zone is not zero-rated.

14. Suggested Worker Commands

PLM-CFG-07 is mostly API-driven. Workers are optional and can be consolidated into catalog workers.

```
plm:voice-prefix-import
plm:voice-rate-import
plm:voice-tariff-validate-future-activations
plm:voice-tariff-cache-warm
```

Recommended schedule:

Command	Frequency
plm:voice-prefix-import	Event-driven when large import is uploaded.
plm:voice-rate-import	Event-driven when large import is uploaded.
plm:voice-tariff-validate-future-activations	Daily.
plm:voice-tariff-cache-warm	On deployment and after tariff change events.

15. Minimum Implementation Order

Build in this order:

1. Migrations for tariff plan, zone, prefix, time band, rate, allowance, binding.
2. CRUD APIs for zones and time bands.
3. CRUD APIs for tariff plans and rates.
4. Prefix longest-match resolver.
5. Binding resolver.
6. Time band resolver.
7. Rate overlap validation.
8. Rating lookup API.
9. Activation/retirement API.
10. Events for cache invalidation.
11. Bulk import for prefixes.
12. Bulk import for rates.
13. Backoffice screens.

16. Test Scenarios

16.1 Mobile Call Lookup

Input:

- Called number +254711222333.
- Package pkg_INET_VOICE_STD.

Expected:

- Prefix +2547 matches.
- Zone KE_MOBILE.
- Rate 3.00 KES/minute.

16.2 Longest Prefix Wins

Input:

- Prefixes +2547 and +254711.

- Called number +254711222333.

Expected:

- +254711 wins.

16.3 Emergency Call

Input:

- Called number 999.

Expected:

- Zone EMERGENCY.
- Charge policy ZERO_RATED.
- Rate 0.

16.4 Unknown Destination

Input:

- Called number has no matching prefix.

Expected:

- Lookup returns VOICE_DESTINATION_ZONE_NOT_FOUND.
- RAT-01 quarantines the CDR.

16.5 Historical Rate Lookup

Input:

- Old rate active until 2026-06-30.
- New rate active from 2026-07-01.
- CDR started 2026-06-15.

Expected:

- Old rate is returned.

16.6 Binding Priority

Input:

- Package binding and subscription override both exist.

Expected:

- Subscription override tariff plan wins.

17. Developer Checklist

Before marking this catalog complete:

- Longest-prefix matching is deterministic.
- Rate overlap validation runs before activation.
- Lookup API does not return draft rates.
- Emergency zone validation enforces zero-rated config.
- Premium zones require explicit rates.
- Effective-date logic uses call start time.
- Time band logic uses operator timezone.
- Binding priority is implemented.

- Events are emitted after plan/rate/prefix/binding changes.
- RAT-01 can use lookup API without direct database reads.